

MongoDB SRE Production Audit & Observability Manual

Production Health Check SOP, Audit Playbook & Alerting Guide

AUDIENCE TARGET: SITE RELIABILITY ENGINEERS (SRE) & ON-SITE DBAS

1. Site Reliability Engineer (SRE) Production Audit SOP

This Standard Operating Procedure (SOP) guides SREs and On-Site Engineers in running automated discovery, configuration auditing, and observability verification on live production MongoDB sharded clusters.

Safety Guarantee: All audit operations are read-only diagnostic checks and do not acquire exclusive write locks or alter data.

2. Executing the Automated Audit Suite

Run the Ansible Playbook or Python Discovery script directly against the production target:

```
# Option A: Via Ansible Playbook
ansible-playbook -i inventory.ini audit_mongodb.yml

# Option B: Direct Python Execution
python3 mongodb_cluster_audit.py mongodb://production-mongos:27017
```

The audit script outputs two reports:

- 1. mongodb_audit_report.json (Machine-readable AI telemetry)
- 2. mongodb_audit_report.html (Human-readable executive & SRE report)

3. Health Check Rules & Production Metrics

Check ID	Target Metric / Parameter	Optimal Threshold / Rule
CHK_QUORUM	3-Node Voting Member Quorum	PASS if 3 voting members per set
CHK_OPLOG	Oplog Retention Window	WARN if < 6 hours window
CHK_SPLIT	Global Auto-Split State	WARN if enabled under heavy write load
CHK_BALANCER	Active Balancer Window	WARN if unconstrained 24/7
CHK_BACKUP	PBM PITR Streaming Status	CRITICAL if stream stalled > 5 mins
CHK_CACHE	WiredTiger Cache Pressure	WARN if cache usage > 85%